

# Towards a Reengineering of Business Process Management Lifecycle by the Use of Social Networks

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**Abstract:** The continuous process improvement(CPI) needs to take into account the clients feedbacks in order to have more performance in service delivery and quality of production. This paper presents the first elements of a new solution to use the social networks data by measuring key performance indicators (KPI) to improve the quality of business processes(BP). In this work is described the first step to modify the classic BPM lifecycle by adding the SN data.

**Keywords :** business process management, social network, business process, lifecycle, KPI.

## I. Introduction

Nowadays, companies come closer to the customer (survey, web portal, customer service, ...) to improve the quality of their products and therefore the quality of their business processes. Huge masses of information are available in social networks (SN) without being exploited. One of the ways that the enterprises are using social software is the integration of customers into product development by using blogs to capture ideas for new products and features [1]. Nowadays, SN software provides new opportunities for the design of business processes through the involvement of all stakeholders [2]. The addition of the social dimension in BPM will allow the customer and end-user to participate in the design and implementation of business processes [3], which reduces the rigidity of traditional BPM systems where activities and stakeholders are fixed previously. The use of external resources such as SN will collect information that may come from customers, competitors, and suppliers to improve business processes.

Some works in the research of improving the BPM life cycle tried to use social software as a mean to enhance collaboration between actors without using external resources to the BPM systems as the knowledge within the SN itself. The researches about using SN to improve the BPM performance can be classified in two categories according to the nature of the SN. The SN can be inside the organization like SN defined by relation between employees, or outside like SN on Facebook, Twitter, blogs, etc. The first category ([4], [5], [6], and [7]) has used SN to improve collaboration and knowledge management capitalization. The second category ([8], [9] and [10]), can have more perspectives to improve BPM at design and execution of business processes. Although several research techniques are developed in the mining of SN, they are not deployed in real applications. Current research in the SN are focused on the data acquisition and preparation, anonymization, community structure, dynamics of networks,

expert-finding methods, assembling teams of experts, influence maximization, and information propagation through SN [11]. The use of a technique or another for a business depends on its nature: operational process, or management process as defined by the APQC [12]. For each type, there are: a process category, and process activities, and for each activity the more effective SN technique. This work tries to improve the diagnosis of business processes by the use of public SN data as external sources of knowledge to the organization. It tries to enhance the traditional BPM life cycle by the integration of SN.

## II. OUR APPROACH TO USE SOCIAL NETWORKS IN BPM

### A. Communities in Social BPM

[8] defines three communities to social BPM: internal community, community across the value chain and external community. We note here that both external community and community across the value chain are at the outside of the organization. Each organization community can form and / or be present in SN. We are especially interested by communities outside of the organization (front office) and for which we can extract knowledge we do not have any vision.

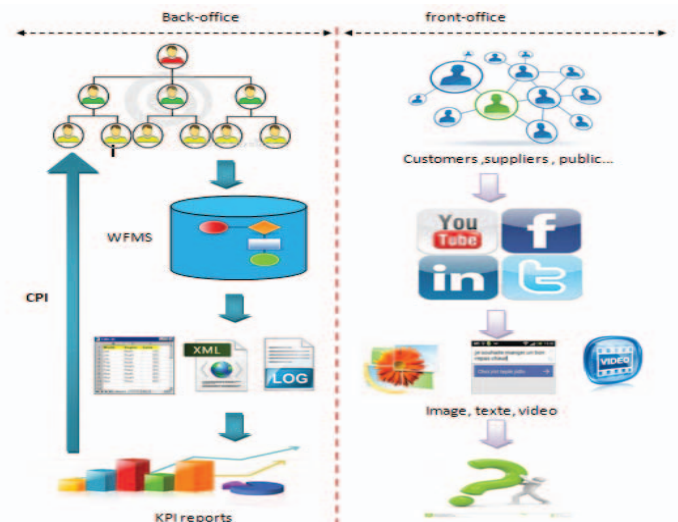


Fig.1: communities and tools in the BPM life cycle inspired by [9]

The Fig.1 shows the tools and communities in the BPM environment. It is clear that SN data coming from the front-office (external community and community across the value chain) are not used yet to enhance the BPM life cycle.

### B. Including SN to enhance BPM lifecycle

Literature offers several life cycles ([13], [14], [8] and [15]) for BPM. In this work we will adopt the life cycle proposed by

[15] for its clarity and its details. This BPM life cycle contains the design, implementation, execution, and evaluation. The diagnosis phase includes the analysis of the current situation to identify the problems due to the method of work and set goals [15] to achieve by design. At this stage, the objectives must be expressed in a qualitative way in the first time, then in a quantitative manner. A qualitative objective can be customer satisfaction, this can be expressed quantitatively through key performance indicators (KPIs) such as the time of making an order. A Key Performance Indicators (KPI) is a metric which is used to determine how an organization is performing against its business objectives. [15] describes KPI as a means to measure the goals in the diagnosis phase. According to him, the continuous improvement of BPM lifecycle requires measurement of objectives during the enactment phase, to compare with the values already defined in the diagnostic phase so-called the null measurement.

We believe that the continuous improvement of the BPM life cycle should begin with the diagnosis phase. It is for this reason that the views of stakeholders (including customers and suppliers), competitors, and even the public at large can provide very helpful information to the organization. This source of information that comes from outside of the organization will be used to measure objectives by the mean of indicators that will be added to the internal indicators considered in some cases subjective because it is often subject to the interpretation of internal actors. Our work aims to use public SN like Facebook, Twitter, etc. Our approach is based on objectives measurement through the use of KPI (Fig.2). In the enactment phase we measure KPIs from the BPMS data like databases and execution logs. These KPI measured by the use of internal resources of the organization will be considered as internal KPI(s). In the other hand, we will use the data collected from public SN to measure other KPI(s) which will be considered as external KPI. The calculated KPI(s) will be used for two kinds of comparison: internal KPI versus external KPI, and correlation between internal and external KPI(s).

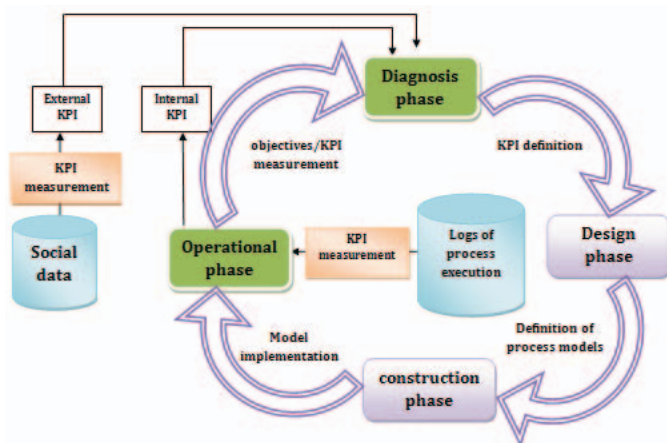


Fig. 2: objective-measurement approach to enhance BPM lifecycle

### III. Conclusion

This work is a first step towards a re-engineering of the classic BPM life cycle in diagnosis phase to improve business processes. This paper presents the core of the idea of

transforming the classic BPM life cycle model by providing adequate place to the SN. Currently, SN contain huge quantity of information that can be a valuable resource for innovation and improvement. We think that including SN data in the BPM lifecycle to enable companies being always informed about their customers, competitors, innovation ideas and follow the market trends in a CPI way. We presented the different communities around the BPM, we think to include them in the BPM lifecycle through the use of their data on SN. The current methods of mining SN can extract information according to the process type, it cannot extract specific information like KPI, it doesn't help organizations measuring their objectives. Our approach is a general approach that can be applied to all types of business processes. The definition of the types of KPI, the extraction methods that will be used and the selection of KPIs cannot be presented in this work because of the limitations of the paper size, although all these points are subject of another works.

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